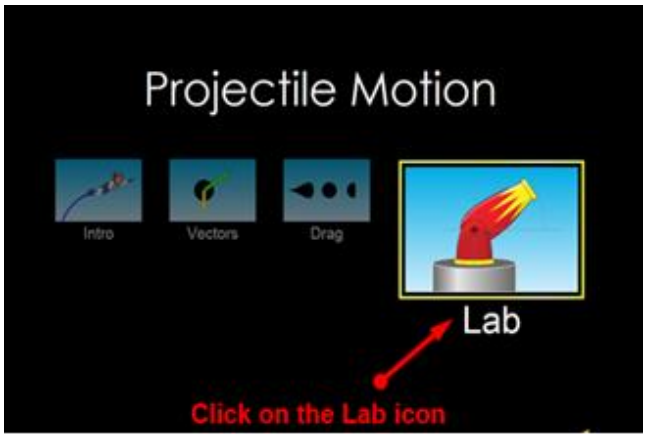
Lab 6: Projectile Motion

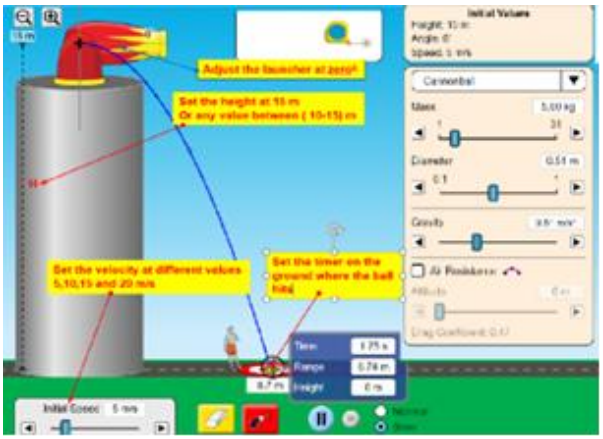
Name: _____

Part 1:

Objective: To show that the time of flight of a horizontal projectile is independent of its initial speed.

1. Open the projectile motion PhET Simulation:
https://phet.colorado.edu/sims/html/projectile-motion/latest/projectile-motion_en.html





2. Adjust the height of the cylindrical base to 10 m by clicking on the **+** and dragging upward.
3. Adjust the angle of the launcher to zero degrees.
4. Set the initial speed of the projectile to 10 m/s.
5. Fire the cannon by pressing the red cannon button at the bottom of the screen.
6. Measure the flight time and range by dragging the Time/Range/Height probe at the top of the screen to the last data point, where the cannonball hit the ground.
7. Repeat the previous steps for four different values of velocity as shown in Table 1 and measure the corresponding time and range. Report your data in Table (1).

Table 1:

Initial Velocity (m/s)	Time (s)	Range (m)
2		
4		
6		
8		
10		

Question 1: How does initial velocity affect the flight time according to Table 1?

Question 2: How does initial velocity affect the range according to Table 1?

Question 3: Does the air resistance change the flight time or the range?

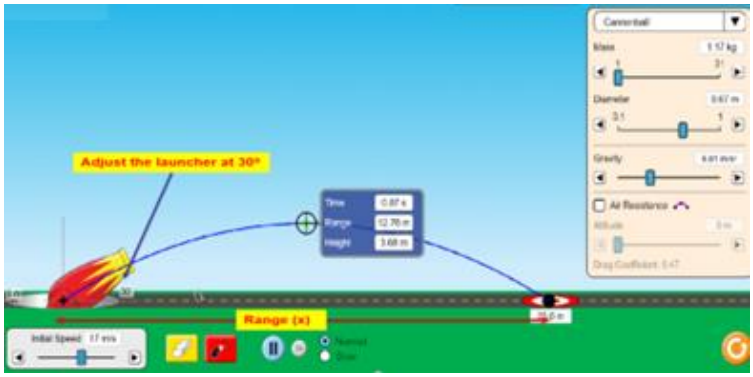
8. Now set the diameter of the cannon ball to its maximum value (1 m) and fire the cannon.

Question 4: Did air resistance change the flight time or range for the larger diameter cannonball? Explain.

Part 2:

Objective: To verify that the horizontal component of the velocity (v_x) of the projectile is constant during its motion.

1. Set the height of the cylindrical base to zero and adjust the launcher to 30° as shown below.



2. Set the initial speed of the projectile to 15m/s and fire the cannonball.
3. Measure the time and range for each data point by dragging the Time/Range/Height probe at the top of the screen to each data point and recording your values in Table 2.
4. Using these time and range values, calculate the average horizontal velocity for each time interval using $v_x = \frac{\Delta x}{\Delta t}$, where Δx is the horizontal distance, or range, and Δt is the time to reach that distance. Record your values in Table 2.

Table 2

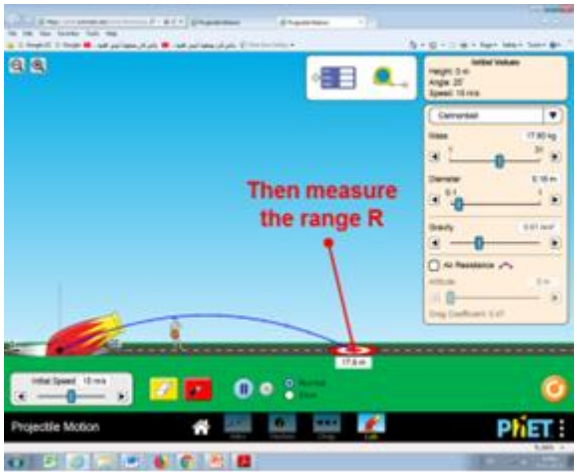
Distance (m)	Time (s)	Horizontal velocity (m/s)

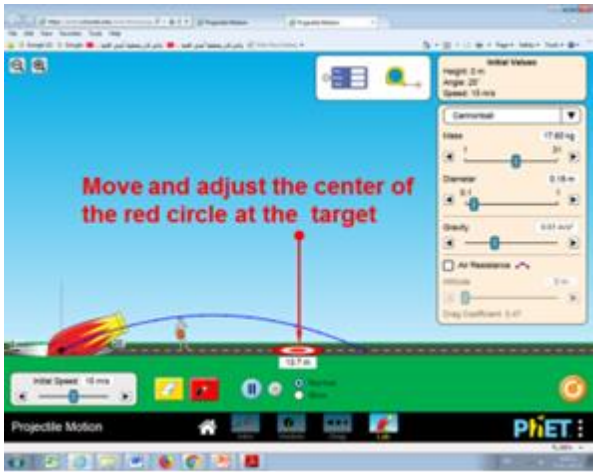
Question 5: Does the horizontal velocity remain constant throughout the entire flight of the projectile?

Part 3:

Objective: To study the relationship between the angle of the projectile and its range.

1. Reset the simulation by clicking on the circular orange reset button on the bottom right of the screen.
2. Adjust the angle of the launcher to 30° and set the initial speed to 15 m/s.
3. Click on the red button to release the projectile and move the red target circle horizontally to where the projectile hit the ground to measure the range R as shown below.





1. Repeat steps 2 and 3 for the angles 30°, 40°, 50°, 60°, 70°, 80° and record the values in Table 3.
2. Calculate sin (2θ) for each angle and record these values in Table 3.

Table 3

v_o= 15 m/s		
Θ (°)	Sin (2θ)	Range (m)
30		
40		
50		
60		
70		
80		

Question 6: Do you see a relationship between initial angle and range? Explain

Question 7: What angle produces the biggest range?

3. Use Google Sheets or Excel to plot Range vs. sin (2θ).
4. Add a linear trendline to this data. Make sure to show the equation of the trendline on the graph and to include a title. The slope of your graph is equal to $\frac{v_0^2}{g}$. Calculate the acceleration due to gravity from your slope.

Question 7: What is your experimental acceleration due to gravity, g_{exp}?

5. Determine the percent error between the experimental value (slope of your graph) and the accepted value for g (9.80 m/s^2).

$$\%error = \frac{|accepted - \exp e\,rimental|}{accepted} \times 100\%$$